

Splunk4Rookies - Platform AI Workshop:

Lab Exercises

No Splunk Expertise Required: Accelerate Time to Insight with AI

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Overview

Welcome to Splunk Gen AI and MCP Workshop.

In today's lab, we're going to explore the power of AI in Splunk using both the AI Assistant and the Splunk MCP Server.

At the end of today's lab, you will have:

- Created SPL, executed SPL and have SPL results explained to you from natural language prompts using the AI Assistant.
 - Explained SPL using the AI Assistant.
 - Learned how to use audit data to understand the queries and responses the AI Assistant generates.
 - Configured the Splunk MCP Server in Splunk Cloud.
 - Configured an MCP client to connect to the Splunk MCP Server.
 - Used your LLM / MCP Client to query Splunk and perform some basic Splunk workflows.
-

Lab #1: Splunk AI Assistant Exercises

Overview

This hands-on lab will demonstrate the Splunk AI Assistant's capabilities to **write, explain, personalize, optimize, run and explain the results from** SPL queries.

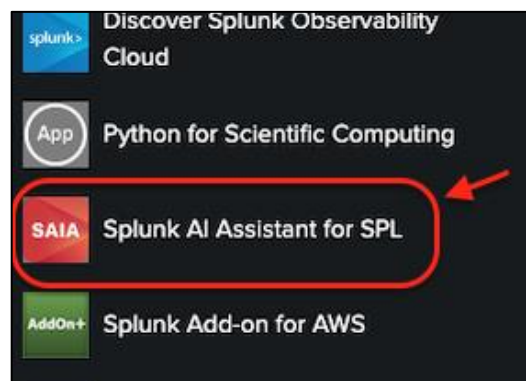
Completing this activity enables you to:

- Demonstrate AI Assistant's core capabilities: **write, explain, optimize, run and explain results from** SPL
 - Use the audit functionality to review how users are interacting with the assistant
-

Connecting to the Lab #1 environment

Log into your Splunk MCP Server Cloud Instance as your associated user credentials supplied in the **Workshop Credentials Spreadsheet**.

Navigate to the “Splunk AI Assistant for SPL” App.



Scenario

You are a new analyst in a SOC team that uses Splunk. Despite being unfamiliar with the platform, you have done your research and identified that the Splunk AI Assistant could be a good means to learn how to get value out of Splunk faster.

Exercise 1: Identify some potentially suspicious emails

Description

There have been some reports of a high volume of emails coming in from gmail accounts and you have been asked to investigate what is going on.

Steps

1. Click the “+ New Chat” button in the upper left corner of the “Splunk AI Assistant” page.

Note that as you perform the following tasks you may be asked to approve the execution of SPL by the assistant, and will see modals like the one in the screenshot below

Approve tool execution

×

Approve running this Splunk search before execution.

Tool

Splunk search

Earliest

-30d

Latest

now

Max events

100

Search

```
search index=main sourcetype=o365:reporting:messagetrace SenderAddress="*@gmail.com" |
stats count as messages values(Status) as statuses min(_time) as firstSeen max(_time) as
lastSeen by SenderAddress | sort - messages | head 20
```

Deny

Approve

Always execute for this session

Also note that because the data generation may differ in your instance and SPL generation is non-deterministic (i.e. will not always be the same even if you ask the same question) your results may differ widely from those in this guide.

2. Think about some of the following questions to help you get started...
 - a. Do we have any log data in our Splunk instance that relates to email?
 - i. Try this prompt: Is there any data in this environment that contains email logs?
 - b. From that log data, do any of the events relate to gmail email addresses?
 - i. Try a prompt like this: From those email log sources can you identify any data that contains senders with a gmail domain?
 - ii. You might then see some results that look like this



From those email log sources can you identify any data that contains senders with a gmail domain?



> **Thought for 13s**

Yes — the O365 message trace logs contain Gmail senders. I queried `index=main sourcetype=o365:reporting:messagetrace` for `SenderAddress` ending in `@gmail.com` over the last 30 days and found matches.

SPL



```
index=main sourcetype=o365:reporting:messagetrace
SenderAddress="*@gmail.com"
| stats count as messages values(Status) as statuses min(_time) as
firstSeen max(_time) as lastSeen by SenderAddress
| sort - messages
| head 20
```

- c. If they do, who are they being sent to?
 - i. Try a prompt like this: Can you show me who the recipients of those emails are?
 - ii. You might see some results like this

Execution Results

• Result rows (sender+recipient pairs): 1

SenderAddress	RecipientAddress	messages	statuses	firstSeen	lastSeen
am6115912@gmail.com	compliance@froth.ly	7764	FilteredAsSpam	1782245998	17824

💡 If you want, I can also show:

- Top recipients overall (ignore sender, just `RecipientAddress`)
- Per-day trend of Gmail-to-recipient volume
- Example subjects being sent to each recipient (use `values(Subject)` / `top Subject`)



- d. Are there any interesting patterns you can identify related to the volume of this email traffic?

- i. Try a prompt like this: Can you show me the volume of emails over time from that sender to that recipient, broken down day-by-day?
- ii. You may see some results like this

 Execution Results (sample table)

• Rows returned (days): 31

_time (day)	FilteredAsSpam
2026-06-23	264
2026-06-24	4412
2026-06-25	3100

 **Note:** The earlier days in the 30-day window show 0 volume; activity spikes start on **2026-06-23**.

If you'd like, I can also return a second column for **total messages per day** (regardless of status) and/or convert this into a simple single-series chart (messages/day) instead of splitting by status.



- e. Can the AI Assistant summarize your findings in a short paragraph to take back to the SOC manager?
 - i. Try a prompt like this: Can you summarize the findings from this chat for me in 200 words or less?
 - ii. You may see some results like this



Can you summarize the findings from this chat for me in 200 words or less?



We confirmed this Splunk environment contains email-flow logs from Microsoft 365 / Office 365 message trace data in `index=main` with `sourcetype=o365:reporting:messagetrace`. Sample events include typical mail-trace fields such as `SenderAddress`, `RecipientAddress`, `Subject`, `Status`, `MessageId`, `MessageTraceId`, `Received`, `FromIP/ToIP`, and `Size`.

We then searched these message trace logs for senders with a Gmail domain (`SenderAddress="*@gmail.com"`) over the last **30 days**. That search returned one Gmail sender: `am6115912@gmail.com`, with approximately **7.7K** messages observed, and the status reported as `FilteredAsSpam`.

Next, we identified the recipient(s) of those Gmail-sender messages. For the same time range, the Gmail sender's messages were addressed to `compliance@froth.ly`.

Finally, we trended volume day-by-day for `am6115912@gmail.com` → `compliance@froth.ly`. The daily volume was **0** for most of the 30-day window, then spiked starting **2026-06-23** (264), **2026-06-24** (4412), and **2026-06-25** (3100), all categorized as `FilteredAsSpam`.

3. Great job, you have identified the spike of email traffic from gmail accounts, and also have a nice report to give back to your manager!

Exercise 2: Support the Detection Engineering Team

Description

Your manager has asked you to spend a few days with the detection engineering team to get more familiar with how they work. In honesty though, you are struggling with some of the complex searches they have been writing and have started to wonder if the AI Assistant could help demystify things for you.

Steps

1. Click the “+ New Chat” button in the upper left corner of the “Splunk AI Assistant” page.
2. You have been handed a particularly complex detection rule and asked to make sure it is included in the Detection Engineers knowledge base, with a description of what it is doing.


```

| tstats `security_content_summariesonly` count min(_time) as firstTime max(_time) as
lastTime
from datamodel=Web where
Web.http_user_agent =*
NOT Web.http_user_agent IN (
    "-",
    "unknown"
)
by Web.http_user_agent Web.dest Web.src Web.status
| `security_content_ctime(firstTime)`
| `security_content_ctime(lastTime)`
| `drop_dm_object_name(Web)`
| eval http_user_agent = lower(http_user_agent)
| lookup scripting_tools_user_agents tool_user_agent AS http_user_agent OUTPUT tool
| where isnotnull(tool)
| stats count
    min(firstTime) as first_seen
    max(lastTime) as last_seen
    values(tool) as tool
by http_user_agent dest src status
| `security_content_ctime(first_seen)`
| `security_content_ctime(last_seen)`
| `http_scripting_tool_user_agent_filter`

```

3. You're going to use the AI Assistant to help you describe this detection rule:
 - a. Try a prompt like this: Please help describe what this SPL is doing <inserting the SPL here>
 - b. You may see a result like this



>  Thought for 3s

 **Query Overview**

This search uses the **Web data model** (via `tstats`) to find HTTP events whose **User-Agent string matches known scripting/automation tools** (based on a lookup). It then summarizes the activity by **user agent + src/dest + HTTP status**, including **first/last seen times** and which tool(s) matched.

In other words: it's a "**web requests made by scripting tools (curl/python/etc.)**" detector, using user-agent matching.

 **Component Breakdown (In Order)**

- 1) Fast, summarized retrieval from the Web data model (`tstats`)

4. You spotted that the macros come from Splunk's ESCU repo, maybe you can find out what the link is
 - a. Try a prompt like this: Please help me understand where this detection rule came from, is it from Splunk's ESCU content?
 - b. You may see a result like this

 Please help me understand where this detection rule came from, is it from Splunk's ESCU content?

 > ✨ **Thought for 5s**

Yes — the structure strongly suggests it originated from **Splunk's ESCU / Splunk Security Content** detections.

🔗 **Why this looks like ESCU content**

Your SPL contains multiple “Security Content” macros and patterns that are characteristic of ESCU detections:

- ``security_content_summariesonly``
Commonly used in ESCU to force/encourage use of **data model acceleration / summaries** with `tstats`.
- ``security_content_ctime(...)``

- c. Also check out some of the recommended checks the assistant makes at the end of the result (if any)
5. The results keep referring to macros as well, but you aren't sure what they are
 - a. Try a prompt like this: Please help me understand what macros are, how to use them and how Splunk stores them
 - b. You may see a result like this



Please help me understand what macros are, how to use them and how Splunk stores them



> ✨ Thought for 3s

🎯 What macros are (in Splunk SPL)

A **search macro** is a **reusable snippet of SPL** that you can insert into other searches. Think of it like a “function” or “named shortcut” for SPL.

Macros are used to:

- Standardize filters (e.g., “exclude known-good hosts”)
- Hide complex SPL behind a simple name
- Make detections easier to maintain (update the macro once, all searches using it inherit the change)
- Parameterize common logic (macros can take arguments)

Splunk denotes a macro invocation with **backticks**: `macro_name`.

Docs: [Define search macros in Settings](#)

6. Finally, you want a nice summary so you can input something into your knowledge base
 - a. Try a prompt like this: Please summarize the findings of this chat so I can document this SPL search in my knowledge base
 - b. You may see a result like this



Please summarise the findings of this chat so I can document this SPL search in my knowledge base



🎯 Summary for Knowledge Base: “HTTP Scripting Tool User-Agent” SPL

Purpose / What this search detects

This SPL is a detection-style search that identifies **web requests whose HTTP User-Agent matches known scripting/automation tools** (for example curl-like or programmatic clients), then summarizes the activity by **source, destination, HTTP status, and user agent**, including **first/last seen timestamps** and the **matched tool name**.

⚙️ How the SPL works (high level flow)

1. Fast retrieval from the Web data model

Exercise 3: Help a friend

Description

The detection engineering team have asked you to look at this search that was given to them by one of the SREs, who doesn't use Splunk so often and has asked for some help.

```
index=_internal (host="*") (sourcetype=aws:s3:log OR
sourcetype=aws:logs:log OR sourcetype=aws:sqsbaseds3:log OR
sourcetype=aws:metadata*:log OR sourcetype=aws:cloudwatch:log OR
sourcetype=aws:cloudwatchlogs:log OR sourcetype=aws:kinesis:log)
(datainput="*") level=ERROR
| fillnull value="" ErrorCode, ErrorDetail
| eval ErrorDetail = if((ErrorDetail == "" or ErrorDetail == ""))
and !isnull(message), message, ErrorDetail)
| table count by _time, message, host, sourcetype, datainput,
ErrorCode, ErrorDetail
| eval error_count=1
| lookup aws_health_error_type_lookup ErrorCode, ErrorDetail,
sourcetype OUTPUT ErrorType | rename error_count as count
| lookup aws_log_sourcetype_modinput_lookup sourcetype OUTPUT modinput
| timechart sum(count) by modinput
```

Can the assistant help you explain this search? Are you able to identify any areas where the SRE could improve the search?

Steps

1. Notice that the query is reasonably complex, try running from the search bar:

New Search Save As ▾

```
index=** (host="**") (sourcetype=aws:s3:log OR sourcetype=aws:logs:log OR sourcetype=aws:sqsbaseds3:log OR sourcetype=aws:metadata*:log OR sourcetype=aws:cloudwatch:log OR
sourcetype=aws:cloudwatchlogs:log OR sourcetype=aws:kinesis:log) (datainput="**") level=ERROR
| fillnull value="" ErrorCode, ErrorDetail
| eval ErrorDetail = if((ErrorDetail == "" or ErrorDetail == '') and !isnull(message), message, ErrorDetail)
| table count by _time, message, host, sourcetype, datainput, ErrorCode, ErrorDetail
| eval error_count=1
| lookup aws_health_error_type_lookup ErrorCode, ErrorDetail, sourcetype OUTPUT ErrorType
| rename error_count as count
| lookup aws_log_sourcetype_modinput_lookup sourcetype OUTPUT modinput
| timechart sum(count) by modinput
```

❗ Error in 'lookup' command: Could not construct lookup 'aws_health_error_type_lookup, ErrorCode, ErrorDetail, sourcetype, OUTPUT, ErrorType'. See search.log for more details.

❗ The lookup table 'aws_health_error_type_lookup' does not exist or is not available.

❗ The search job has failed due to an error. You may be able view the job in the [Job Inspector](#).

No Event Sampling ▾

2. Note that it doesn't seem to be working – looks like you don't have the right permissions to access the data or the apps... How are you going to troubleshoot this search without the right permissions?
3. Head back to the AI Assistant and start a new chat by clicking the “+ New Chat” button in the upper left corner of the Splunk AI Assistant for SPL page.
4. Paste the query you copied above into the prompt field and submit the query to the AI Assistant.

Splunk AI Assistant for SPL

+ New chat

Today

- Aws error log analysis
- New chat
- Admin actions by count
- Windows failed login query

Usage guidelines

Settings

Explain this SPL:

```
index=,internal (host="**") (sourcetype=aws:s3:log OR sourcetype=aws:logs:log OR sourcetype=aws:sqsbaseds3:log OR sourcetype=aws:metadata*:log OR
sourcetype=aws:cloudwatch:log OR sourcetype=aws:cloudwatchlogs:log OR sourcetype=aws:kinesis:log) (datainput="**") level=ERROR
| fillnull value="" ErrorCode, ErrorDetail
| eval ErrorDetail = if((ErrorDetail == "" or ErrorDetail == '') and !isnull(message), message, ErrorDetail)
| table count by _time, message, host, sourcetype, datainput, ErrorCode, ErrorDetail
| eval error_count=1
| lookup aws_health_error_type_lookup ErrorCode, ErrorDetail, sourcetype OUTPUT ErrorType
| rename error_count as count
| lookup aws_log_sourcetype_modinput_lookup sourcetype OUTPUT modinput
| timechart sum(count) by modinput
```

> Ran get_personalization_data ✓

Explanation of the SPL

SPL

```
index=,internal (host="**") (sourcetype=aws:s3:log OR sourcetype=aws:logs:log
OR sourcetype=aws:sqsbaseds3:log OR sourcetype=aws:metadata*:log OR
sourcetype=aws:cloudwatch:log OR sourcetype=aws:cloudwatchlogs:log OR
sourcetype=aws:kinesis:log) (datainput="**") level=ERROR
| fillnull value="" ErrorCode, ErrorDetail
| eval ErrorDetail = if((ErrorDetail == "" or ErrorDetail == '') and
!isnull(message), message, ErrorDetail)
| table count by _time, message, host, sourcetype, datainput, ErrorCode,
ErrorDetail
| eval error_count=1
| lookup aws_health_error_type_lookup ErrorCode, ErrorDetail, sourcetype
OUTPUT ErrorType
| rename error_count as count
| lookup aws_log_sourcetype_modinput_lookup sourcetype OUTPUT modinput
| timechart sum(count) by modinput
```

Step-by-step breakdown

1. Base search

SPL

Results from GenAI can vary; review for accuracy. [View AI details](#)

5. After you paste the query into the prompt field, the Assistant will review the query, break it down into its component parts, and then provide a detailed explanation of the SPL query. Does it catch anything unusual?

Exercise 4: You've been caught using AI!

Description

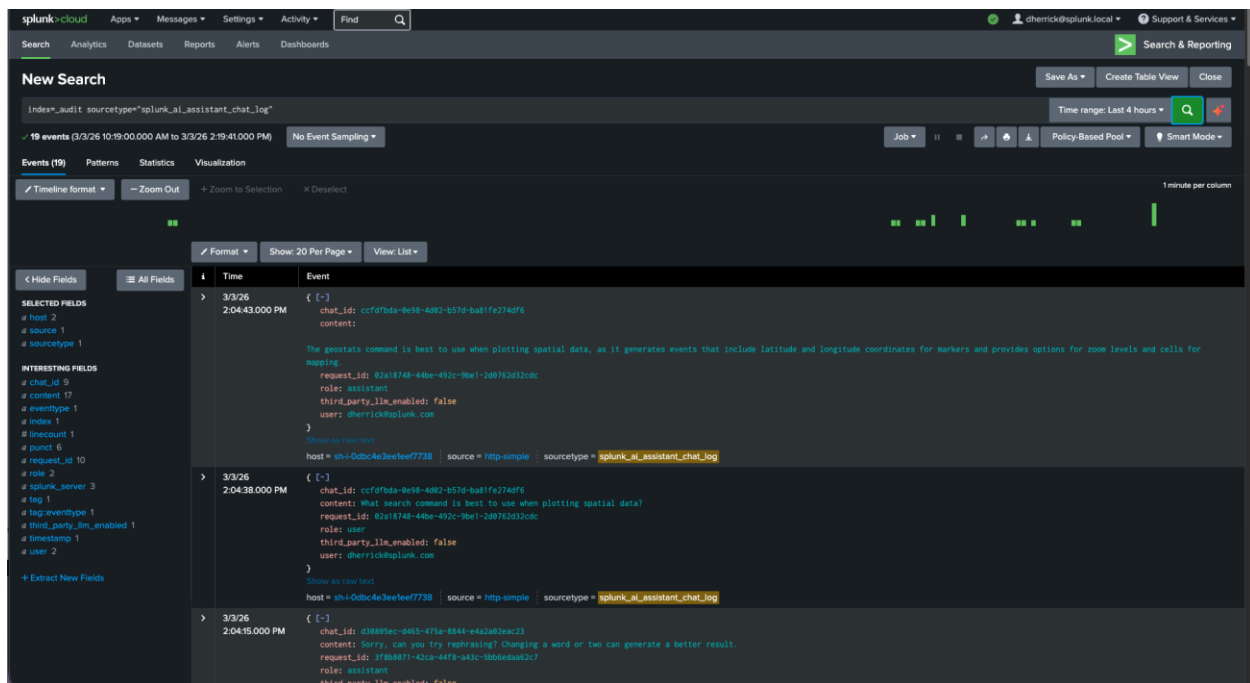
After being praised for your great work you confess to your manager that it was thanks to AI. Instead of being annoyed, they are impressed with your initiative and use of AI! Now they want to know what you have done so the rest of the team can learn from you...

Steps

1. See if you can summarize your actions based on your chat histories. This time it's not about the results of your chat, but more about the way you have structured your questions to find the information you needed.
2. For admin users everyone's chat history is available! They can go to the Search and Reporting app, and enter the following SPL query:

```
index=_audit sourcetype=splunk_ai_assistant_chat_log
```

3. The search will return all the conversations for every user in the given timeframe, as shown below. Note that you will not necessarily have access to the _internal index for the lab exercises, with this search not returning any results (it's something to share with your Splunk admin team!)



Lab #2: Splunk MCP Server Exercises

Overview

In this exercise, you will learn how the Splunk MCP server enables users to access Splunk data from outside the Splunk interface.

When you complete this exercise, you will have:

- Create an MCP server token.
- Configure the MCP Client to connect to and communicate with the Splunk MCP server.
- Verify the connectivity by listing the MCP server tools.
- Test the performance of the MCP server and see how it can present Splunk data to your MCP client.

Exercise 1: Create an MCP Server Token

Overview:

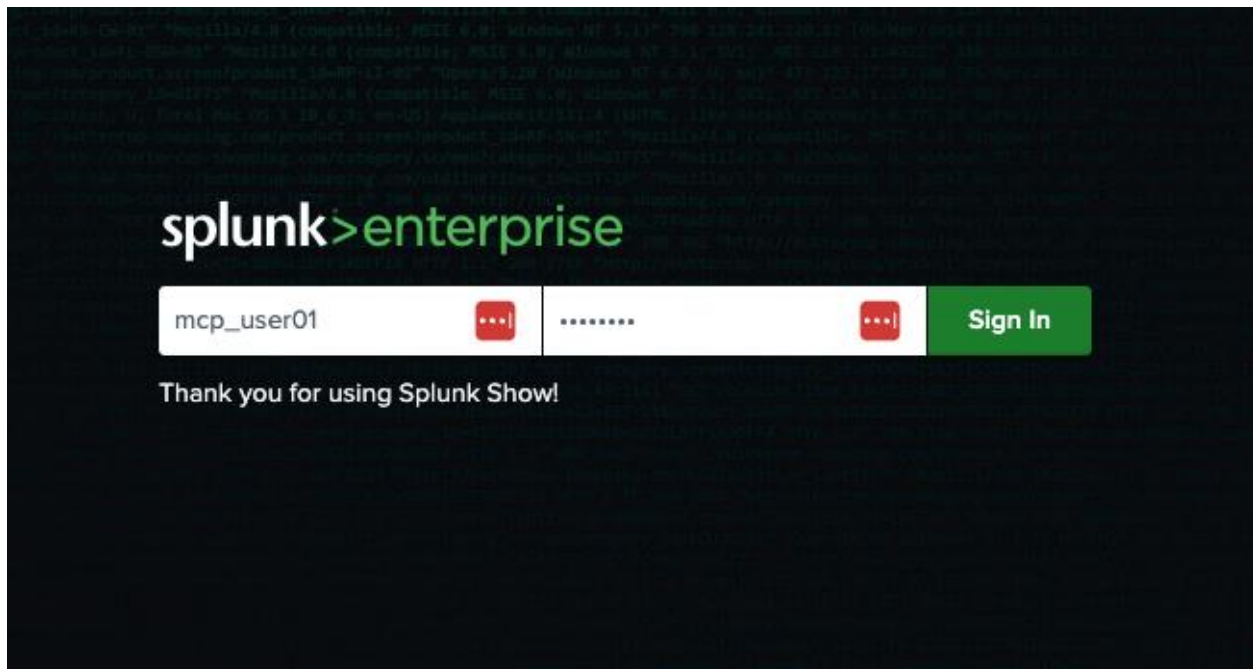
Workshop attendees will now log into Splunk, create a token which will be used for MCP Client access to the MCP Server in Splunk.

Tokens will then be used in the MCP Server Configuration file in your MCP Client.

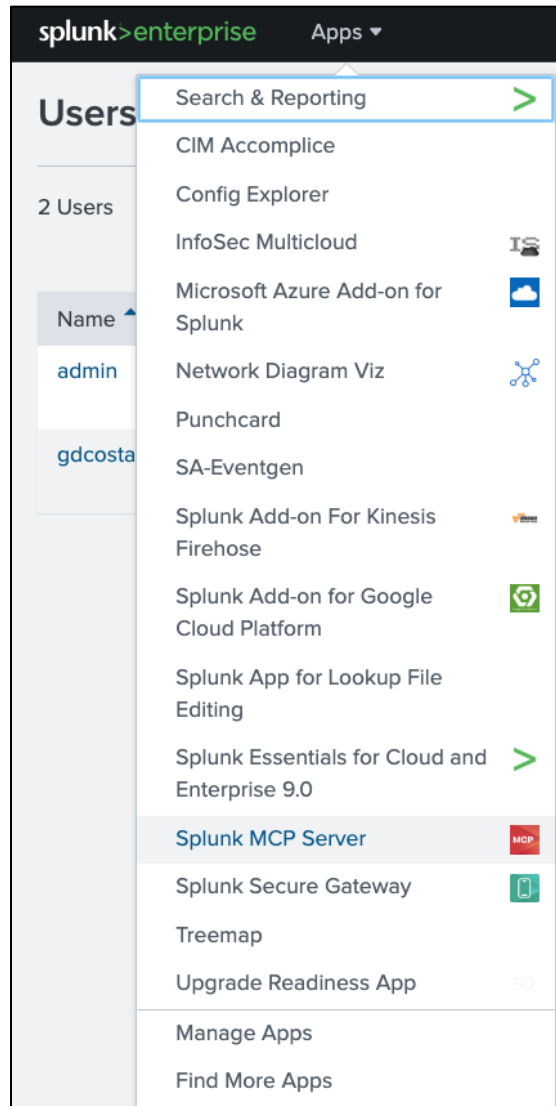
Connecting to the Lab #2 environment

Log into your Splunk MCP Server Cloud Instance as your associated user credentials supplied in the **Workshop Credentials Spreadsheet**.

- Log into Splunk as mcp_userXX



- Click on Apps → Splunk MCP Server



- Click “Create MCP Encrypted Token”
 - In the [User] field, type in the user you just created.
 - Ensure [Audience] is set to “mcp”.
 - For [Expiration] keep “Relative Time” selected with a “+30d”
 - Click Create.
 - An MCP Encrypted Token will be created which you will need for your MCP client.
- Click on the “Copy” button to copy the MCP Encrypted Token.

Create MCP Encrypted Token

User*

mcp_user01

Audience

mcp

Audience is fixed to mcp.

Expiration

Relative Time

+30d

Examples: +10m, +20h, +30d

Not Before

Relative Time

Examples: +10m, +20h, +30d

MCP Encrypted Token

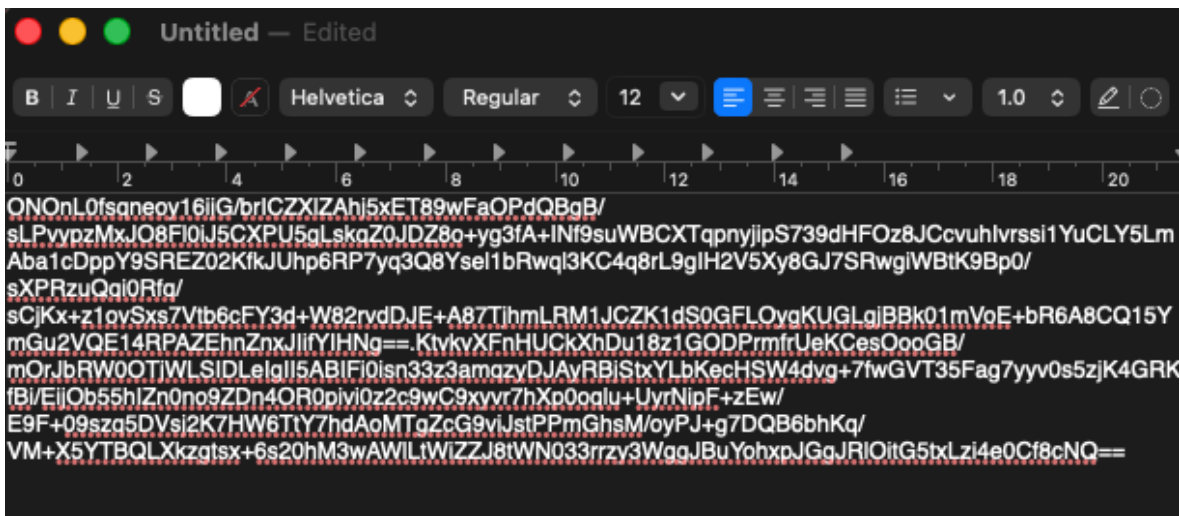
bWUBPnqcqyPmobf7CGpO6xOTljHZ+vjFRHzJN3AotKPofolLpjJjHZtJ910xWynCdETD7Oq/93VVALg/DPAwpTxyMqgssHguM/3H+Gxwh9h2dXI4PfrbM+AtzaMmdFzoG+E+ndt/bdqFVQsDs0cirxc+qBQrk9Ts3hvolfgUEFGcVp12pwD53V5gB6DHgcgt1YTXIL4B/yStDJ5GiTFeEDPFx9IBZyRc7tZCt98SRE5CvwpFUhdgOkYXqsxhRA+7NGBD2r1tDHIQCIQG5i6CHtpoFkEGHAJMnCun86kNXBW89cj+oep7qLOem23AQAT0BJsHMYxdfNSTmKjU1D7flw==.kAaN7ggynl70Y8Cnn4JyACQC

Token is no longer accessible after you close this window.

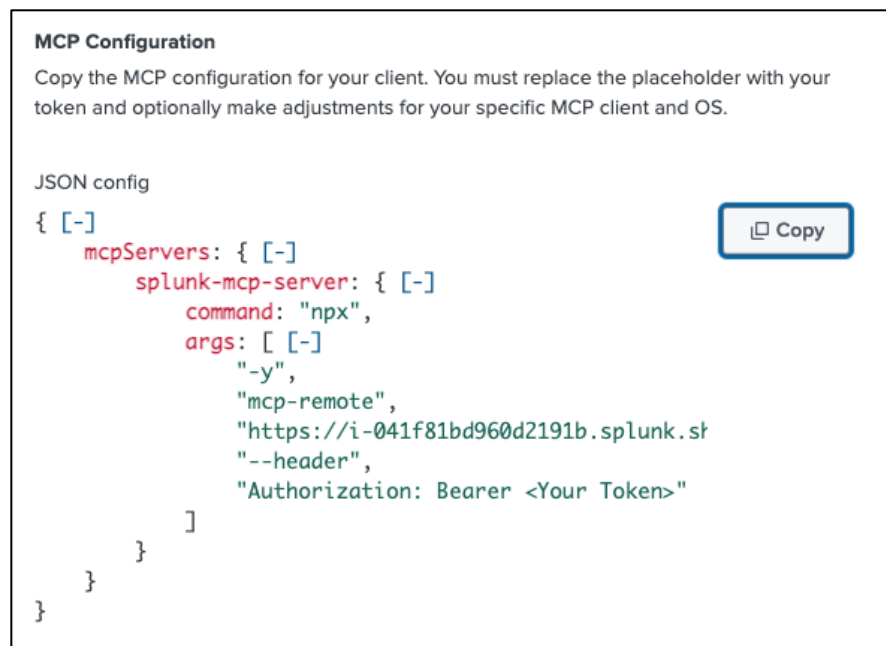
Copy

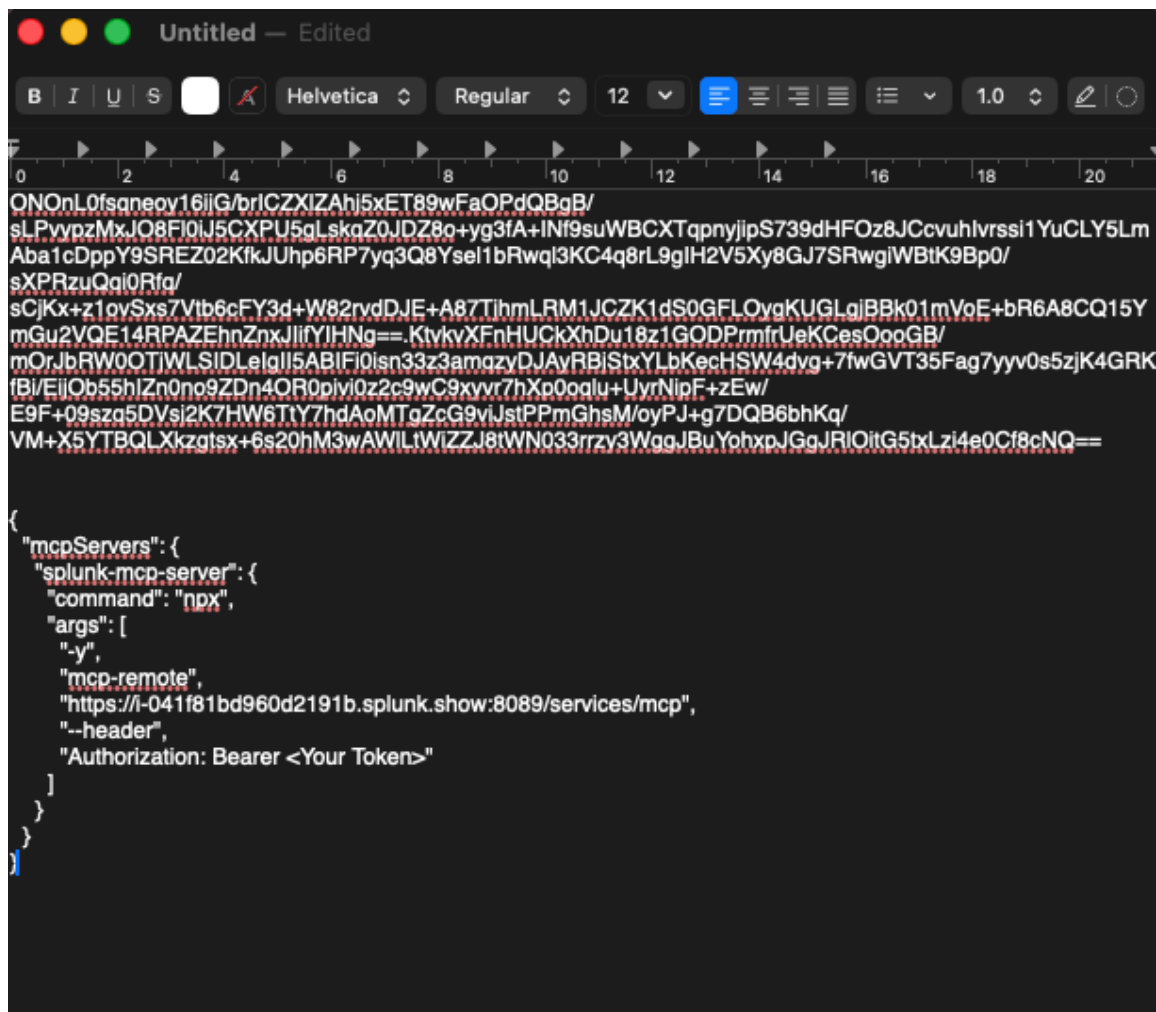
Close

- Paste the token it into a notepad (Windows) or TextEdit (Mac) document – you will need it later.
 - **IMPORTANT:** This will be the **ONLY** time the MCP Encrypted Token will be visible.



- Click “Close”
- Next, copy the “MCP Configuration” and paste it in the same notepad (Windows) or TextEdit (Mac) or another editor you used previously.





```
ONOnL0fsqneov16ijG/brICZXIZAhj5xET89wFaOPdQBqB/  
sLPvypzMxJO8FI0IJ5CXPu5qLskaZ0JDZ8o+yg3fA+INf9suWBCXTqpnYjlpS739dHFOz8JCcvuhIvrssi1YuCLY5Lm  
Aba1cDppY9SREZ02KfkJUhp6RP7yq3Q8Ysel1bRwql3KC4q8rL9gIH2V5Xy8GJ7SRwgiWBtK9Bp0/  
sXPRzuQqi0Rfa/  
sCjKx+z1ovSxs7Vtb6cFY3d+W82rvdDJE+A87TihmLRM1JCZK1dS0GFLOvgKUGLajBBk01mVoE+bR6A8CQ15Y  
mGu2VQE14RPAZEhnZnxJliFYIHNg==.KtkvXFhHUCkXhDu18z1GODPrmfrUeKCesOooGB/  
mOrJbRW00TJWLsIDLeIgl5ABIFi0isn33z3amazyDJAYRBjStxYLBKechHSW4dvg+7fwGVT35Fag7yyv0s5zjK4GRK  
fBi/EiOb55hZn0no9ZDn4OR0pivi0z2c9wC9xyvr7hXp0oqlu+UyrNipF+zEw/  
E9F+09szq5DVsj2K7HW6TtY7hdAoMTgZcG9viJstPPmGhsM/oyPJ+g7DQB6bhKq/  
VM+X5YTBQLXkzgtsx+6s20hM3wAWILtWIZZJ8tWN033rrzy3WggJBuYohxpJGgJRI0itG5txLzi4e0Cf8cNQ==  
  
{  
  "mcpServers": {  
    "splunk-mcp-server": {  
      "command": "npx",  
      "args": [  
        "-y",  
        "mcp-remote",  
        "https://i-041f81bd960d2191b.splunk.show:8089/services/mcp",  
        "--header",  
        "Authorization: Bearer <Your Token>"  
      ]  
    }  
  }  
}
```

- Replace the “<Your Token>” text in the JSON config with the token you created earlier:
- **NOTE:** for **Windows** users, you will also need to change the “command” key-value pair in this way: (Depending on your direct path to npx.cmd)
 - From: "command": "npx"
 - To: "command": "C:\\\\Progra~1\\\\nodejs\\\\npx.cmd"

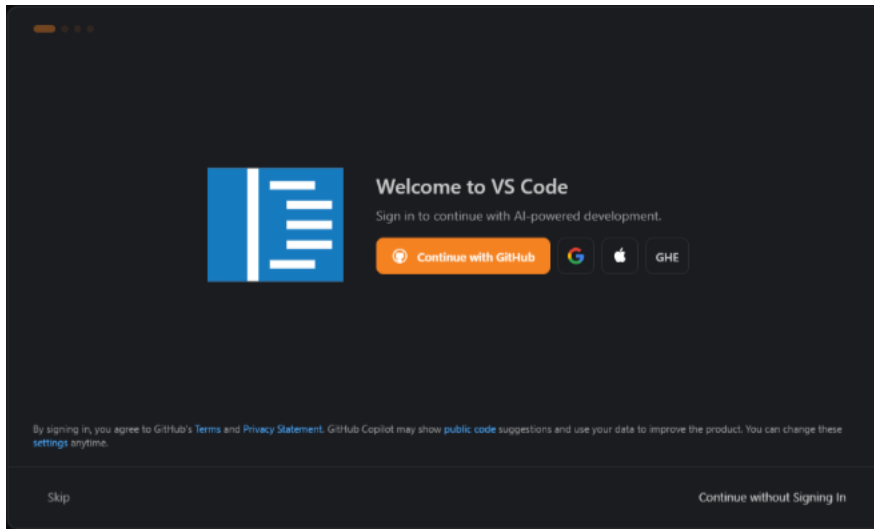
```
{
  "mcpServers": {
    "splunk-mcp-server": {
      "command": "npx",
      "args": [
        "-y",
        "mcp-remote",
        "https://i-041f81bd960d2191b.splunk.show:8089/services/mcp",
        "--header",
        "Authorization: Bearer QONOnL0fsqneoy16ijG/briCZXIZAhj5xET89wFaOPdQBgB/
sLPvypzMXJO8FI0IJ5CXPU5qLskqZ0JDZ8o+yg3fA+INf9suWBCXTqpnjipS739dHFOz8JCcvuhlrssi1YuCLY5Lm
Aba1cDppY9SREZ02KfkJUhp6RP7yq3Q8Ysel1bRwql3KC4q8rL9glH2V5Xy8GJ7SRwgiWBtK9Bp0/
sXPRRzuQqI0Rfq/
sCjKx+z1ovSxs7Vtb6cFY3d+W82rvdDJE+A87TihmLRM1JCZK1dS0GFLOvgKUGLqjBBk01mVoE+bR6A8CQ15Y
mGu2VQE14RPAZEhnZnxJlIfYIHNg==.KivkvXFhHUckXhDu18z1GODPrrmfrUeKCesOooGB/
mOrJbRW00TIWLSIDLelqll5ABIFi0Isn33z3amazyDJAvRBjStxYlbKecHSW4dvg+7fwGVT35Fag7yyv0s5zjK4GRK
fBi/EijOb55hIZn0no9ZDn4OR0piv0z2c9wC9xyvr7hXp0oqlu+UyrNipF+zEw/
E9F+09sazq5DVsl2K7HW6TtY7hdAoMTgZcG9viJstPPmGhsM/oyPJ+g7DQB6bhKq/
VM+X5YTBQLXkzgtssx+6s20hM3wAWILtWlZZJ8tWN033rrzy3WggJBuYohxpJGgJRI0itG5txLzi4e0Cf8cNQ==
"
      ]
    }
  }
}
```

- Save this file locally on your computer.

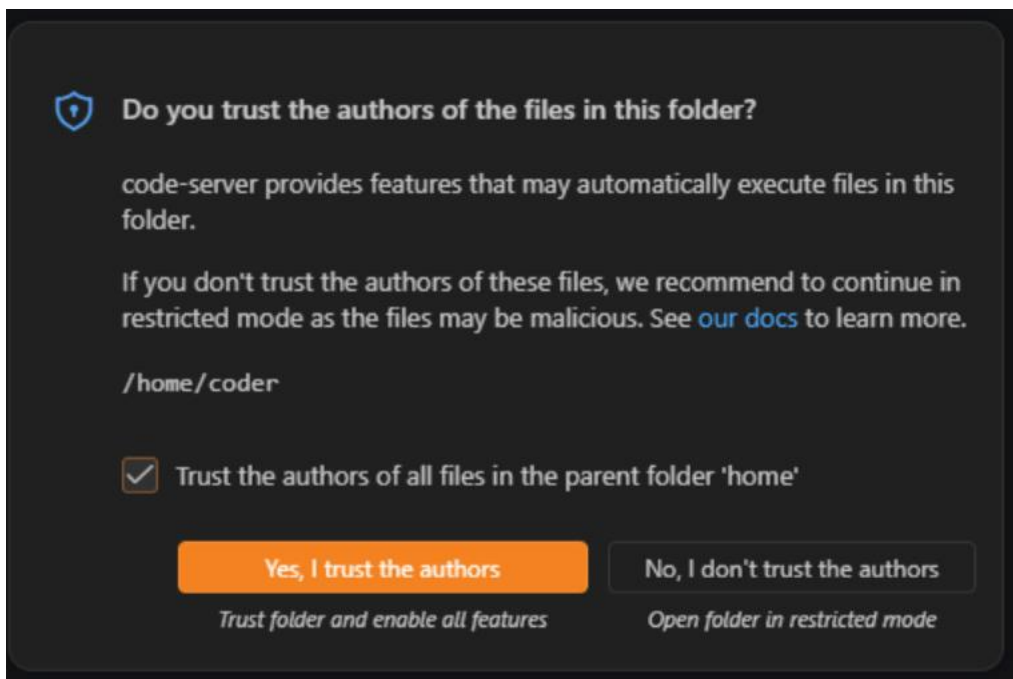
Exercise 2: Configure the MCP Client

A VS Code instance with the Continue extension is pre-provisioned in your lab environment.

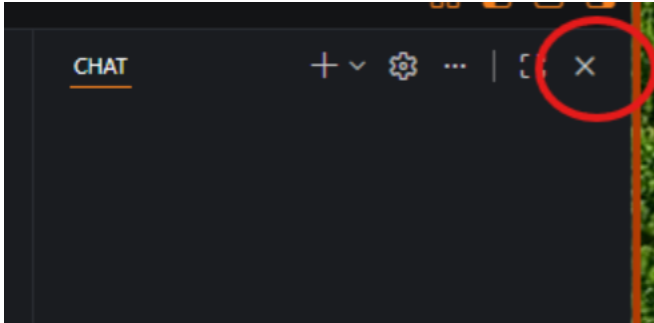
1. Navigate to your VS Code instance in the browser. Click **Continue without Signing in**.



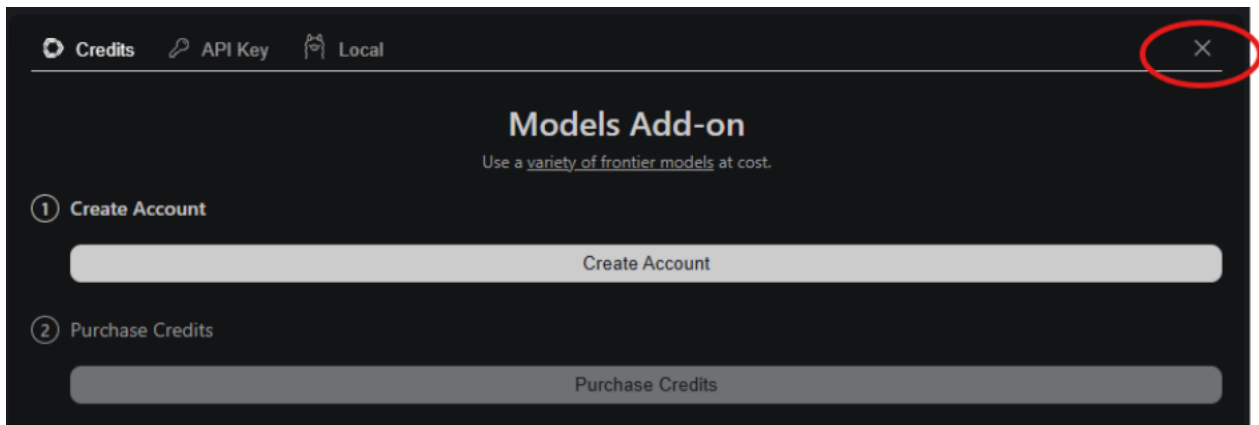
2. When presented with Do you trust the authors of the files in this folder click **Yes, I trust the authors**



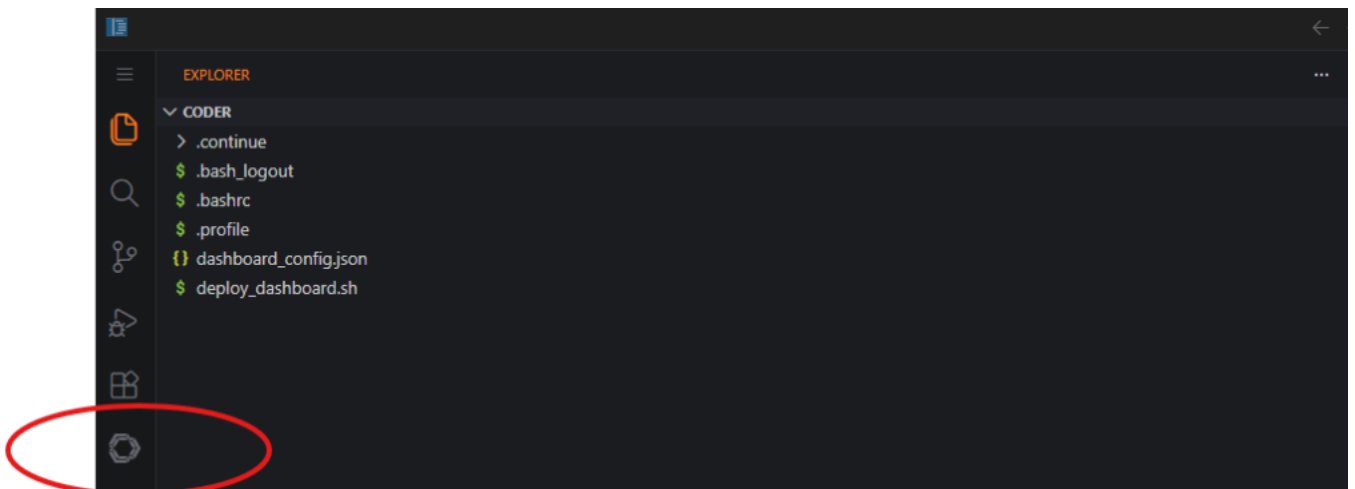
3. If opened, close **Chat** on right side with the **X** button.



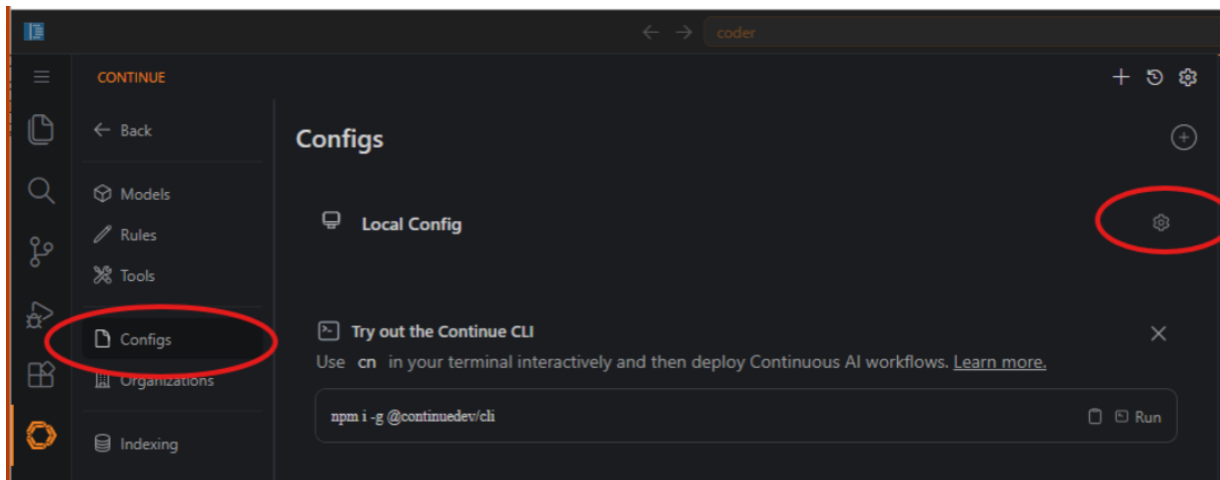
4. If opened, close **the Models** window on the left side with the **X** button.



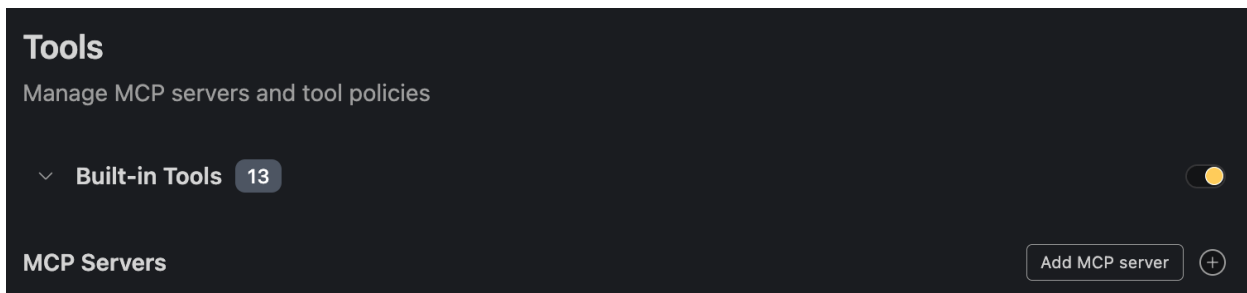
5. You will see VS Code with the Continue extension already installed (look for the Continue icon in the left sidebar — it looks like a speech bubble with a chevron).



6. Open the Continue extension and click the **gear icon** (⚙️), then select **Configs** and then the **gear icon** (⚙️) adjacent to Local Config on the Configs panel. This opens `~/.continue/config.yaml`.



7. This should open up a config showing the AWS Bedrock model connections that are preconfigured for you
8. Now click on tools, where you should see some MCP server and tool policies
9. Click on the + symbol next to the MCP Servers entry to add a new MCP connection



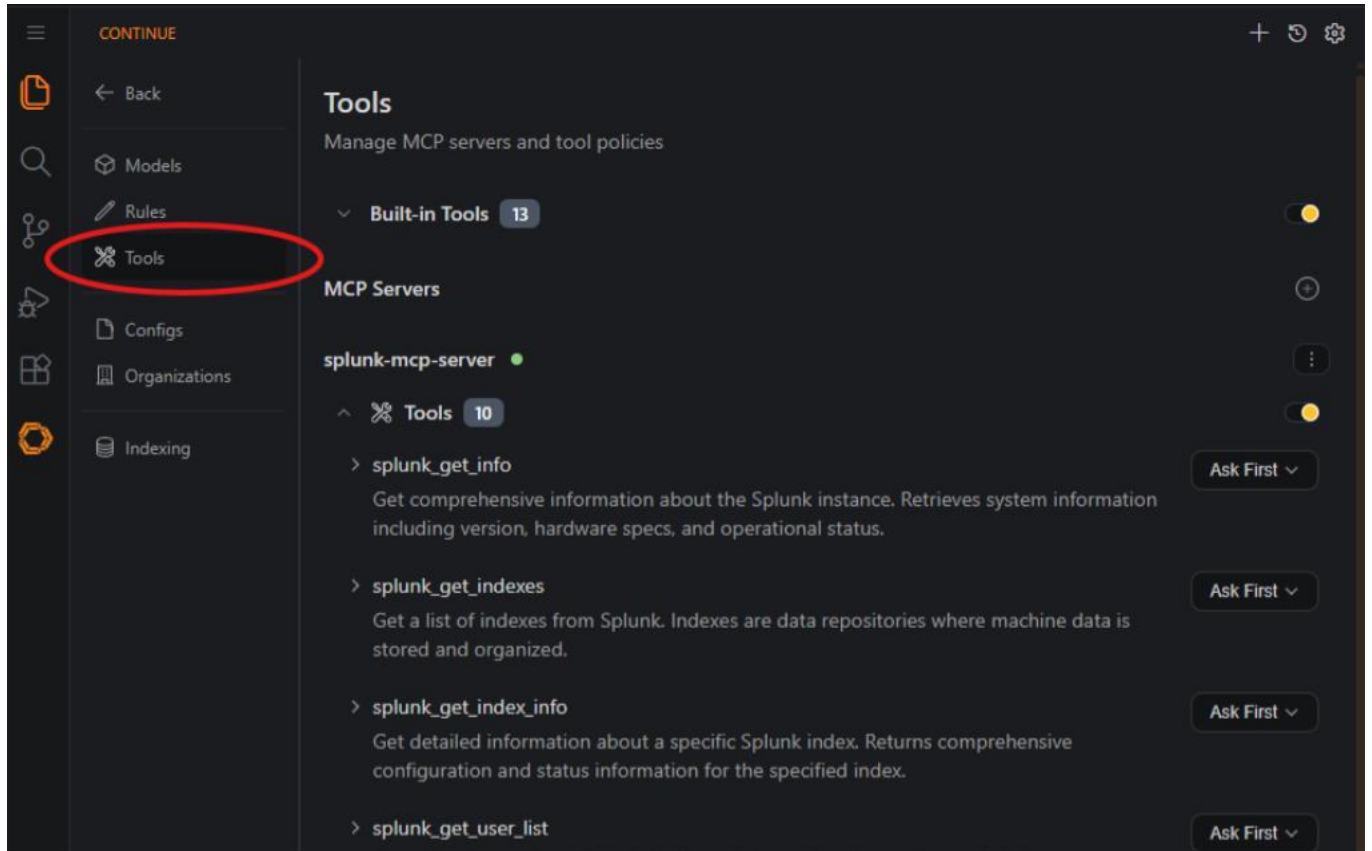
10. Locate the mcpServers section. The server entry is already present. Find the Authorization: Bearer line and replace <YOUR_MCP_TOKEN> with the token you generated above. The full entry should look like this:

```
name: New MCP server
version: 0.0.1
schema: v1
mcpServers:
  - name: New MCP server
    command: npx
    args:
      - "-y"
      - "mcp-remote"
      - "https://<REPLACE_WITH_SPLUNK_INSTANCE>:8089/services/mcp/v1/sse"
      - "--header"
      - "Authorization: Bearer <YOUR_MCP_TOKEN>"
```



```
env:  
  NODE_TLS_REJECT_UNAUTHORIZED: "0"
```

1. Click on **Tools** in the left menu. Verify that you see tools under the splunk-mcp-server.



Exercise 3: Ask Questions of your data using an LLM & Splunk MCP

Now that you've successfully connected your MCP client to the Splunk MCP server, it's time to begin querying your Splunk instance using natural language.

Let's imagine that you are a compliance and risk manager who is trying to understand what the current state looks like with respect to PCI compliance in your organization. You have been given some reports from the business about PCI compliance, but one of your colleagues has told you that log data in Splunk might help you better understand the current compliance.

The problem is that you don't really know how to use Splunk, so you are hoping that your company's VC code assistant can help you figure this out – as apparently it has a connection to Splunk...

Note: you will need to approve actions within the VS code client, and will be prompted to do so throughout the following tasks.

Note: your results may differ from those shown in this guide, due to: different data sources in your environment; and response generation by your MCP client being non-deterministic (meaning your searches and summaries of search results may differ between users and even between running the same question twice!)

Let's start...

1. Given your background is risk and compliance rather than logging, let's start with a few questions to help you figure out what might be useful
 - a. What is the relevance of log data to PCI compliance?

b. What are some example log sources that might be useful to determine PCI compliance?

What is the relevance of log data to PCI compliance?

I can help explain the relevance of log data to PCI compliance. Let me get you comprehensive information about this topic.

<function_calls> </function_calls>

Based on PCI DSS (Payment Card Industry Data Security Standard) requirements, log data is critically relevant to PCI compliance. Here are the key aspects:

Critical Relevance of Log Data to PCI Compliance:

1. Requirement 10: Logging and Monitoring

This is one of the most important PCI DSS requirements, which mandates that organizations must:

- **Log all access** to cardholder data environments (CDE)
- **Track and monitor all access** to network resources and cardholder data
- **Record and retain logs** for a minimum of one year (with at least three months available online)

2. What Must Be Logged:

- **User access activities** - logins, logouts, failed access attempts
- **Administrative actions** - changes to configurations, additions/deletions of users
- **System events** - system crashes, restarts, security event logs
- **Access to cardholder data** - any read, write, or delete operations
- **Invalid access attempts** - authentication failures and suspicious activities
- **Network activity** - connections and disconnections to/from the CDE

2. Now you have a better understand of what logs might be important, let's relate this to Splunk

- a. What log data is available in Splunk?
- b. Can you help me list and document all of the log data in Splunk that is important for PCI compliance?

Can you help me list and document all of the log data in Splunk that is important for PCI compliance?

Absolutely! Let me create a comprehensive inventory of the log data in your Splunk instance that's important for PCI compliance. I'll document this based on the data we've already discovered.

PCI Compliance Log Data Inventory for Your Splunk Instance

1. AUTHENTICATION & ACCESS CONTROL LOGS

Okta Identity Cloud Logs

Attribute	Details
Sourcetype	OktaIM2:log, OktaIM2:user, OktaIM2:groupUser
Host	frothly.okta.com
Event Count	5.2M+ events
Data Categories	User information, Group memberships, Authentication events
PCI Relevance	Requirement 8.2, 8.3, 10.2 - User authentication and access
Retention	Active in Splunk
Status	 CRITICAL

3. Your assistant has identified a few gaps in your logging. Can we come up with a plan to help improve things?
 - a. I see there are some missing data sources, can you outline an outline plan for getting this data into Splunk and explain where some of the data might come from?

I see there are some missing data sources, can you outline an outline plan for getting this data into Splunk and explain where some of the data might come from?

Absolutely! Let me create a comprehensive plan for getting the missing PCI-critical data sources into Splunk. I'll outline the data sources, their origins, collection methods, and implementation steps.

Implementation Plan for Missing PCI Log Data Sources

PHASE 1: DATABASE LOGS (CRITICAL - Priority 1)

1.1 SQL Server Audit Logs

What Data Is Collected:

- User authentication attempts (successful and failed)
- Cardholder data access (SELECT, INSERT, UPDATE, DELETE)
- Administrative changes (schema changes, permissions)
- Stored procedure executions
- Query execution details
- Failed login attempts

Data Origin:

- **Source System:** SQL Server instances in your environment
- **Location:** SQL Server error logs, SQL Server Agent logs, audit files
- **Data Type:** Structured event logs, audit traces

4. OK, so now you know what logs you have you are wondering if you can identify any more detailed issues
 - a. Based on the log data in Splunk, are there any logs that indicate an issue with PCI compliance?
 - b. Can you outline the searches you used just now so I can share them with my Splunk platform team?

Based on the log data in Splunk, are there any logs that indicate an issue with PCI compliance?

Great question! Let me run some searches to analyze the log data in your Splunk instance for potential PCI compliance issues.

Let me check for failed authentication attempts:

Let me check for administrative activity and privilege escalation:

Let me search for suspicious network activity:

Let me check for database and system changes:

Let me check for unusual network traffic patterns:

Let me look for any external data exfiltration attempts:

Let me check for failed Okta authentication events:

✓ Performed 8 actions

✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



5. You've got a load of good insights now, but writing that up to include in your report seems like a bit of a pain...
 - a. Can you write a 500 word paragraph in a report format explaining the findings about PCI compliance in Splunk?

Can you write a 500 word paragraph in a report format explaining the findings about PCI compliance in Splunk?

PCI DSS Compliance Assessment Report - Splunk Log Analysis

Executive Summary of Findings

This assessment analyzed the current state of logging and monitoring capabilities within the organization's Splunk environment to evaluate compliance with the Payment Card Industry Data Security Standard (PCI DSS). The analysis examined 9.6 million events across multiple log sources including authentication systems, network infrastructure, endpoint security, and cloud services. While the organization has implemented foundational logging infrastructure with Okta identity management, Windows system monitoring via Sysmon, pfSense firewall logging, and Microsoft 365 audit trails, significant gaps exist that create material compliance risks and would likely result in major audit findings during a PCI DSS assessment.

The most critical deficiency identified is the complete absence of database access logging, which directly violates PCI DSS Requirement 10.5 mandating comprehensive logging of all access to cardholder data. Organizations cannot demonstrate data protection controls without evidence of who accessed sensitive payment card information, when they accessed it, and what actions they performed. Similarly, payment processing logs from Point of Sale (POS) systems and payment gateways are entirely missing, preventing the organization from maintaining transaction-level audit trails required by Requirement 10.2.5. Web application logs are also absent, leaving no record of application-level access attempts to payment processing functions or cardholder data interfaces.

Additional high-priority gaps include incomplete Windows security event logging, where critical event types such as failed authentication attempts (Event ID 4625), user account creation (Event ID 4720), and object access (Event ID 4663) are not visible in the current dataset. While Sysmon provides valuable endpoint telemetry capturing file creation and process execution, the logs lack sufficient coverage of administrative actions and privilege escalation events necessary for PCI Requirement 10.4 compliance. The Okta authentication logs, though present with 5.2 million events, show limited event type diversity, predominantly featuring policy evaluation challenges rather than comprehensive authentication success/failure tracking, user provisioning events, or privilege grant activities required by Requirement 8.

6. Are there any other steps that you could take based on your interactions with the AI assistant? Anything that the MCP client recommended to you for example?

Exercise 4: Using Splunk MCP Server to support the business

Additional MCP lab exercises, which are dependent on the Infosec Multi-cloud dataset.

Let's imagine that you are a financial analyst working for the CFO, and you have been asked to understand the current state of cloud spend across the business.

Note: you will need to approve actions within the VS code client, and will be prompted to do so throughout the following tasks.

Note: your results may differ from those shown in this guide, due to: different data sources in your environment; and response generation by your MCP client being non-deterministic (meaning your searches and summaries of search results may differ between users and even between running the same question twice!)

1. Let's start by orienting...
 - a. Do you know what data is available in Splunk?
 - b. Is any of this data relevant to your task of understanding cloud spend?
 - c. Do you know which cloud providers you are using?
 - d. Can you determine the total cloud spend across all providers for me?

Can you determine the total cloud spend across all providers for me?

I'll query the billing data across all three cloud providers (AWS, Azure, and GCP) to calculate your total cloud spend.



🔗 📄 ▶ Apply

Let me also get Azure and GCP billing data:

And GCP:

▼ Performed 3 actions

✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



✓ Continue used the New MCP server splunk_run_query tool



2. Now your CFO would like to see a more detailed breakdown of spend by account and by service
 - a. Can you break down the spend on a day-by-day basis for each of GCP, AWS and Azure, also detailing the accounts for each provider?
 - b. Which accounts generated the highest spend?
 - c. Can you break down the spend on a day-by-day basis for each type of service and each cloud provider?

Could you break down the spend by services and identify the services that are incurring the highest charges

Let me query the billing data to break down spending by services for each provider:

✓ Performed 3 actions

- ✓ Continue used the New MCP server splunk_run_query tool 
- ✓ Continue used the New MCP server splunk_run_query tool 
- ✓ Continue used the New MCP server splunk_run_query tool 

Excellent! Now let me get more detailed breakdown with percentage and trend information:

✓ Performed 3 actions

- ✓ Continue used the New MCP server splunk_run_query tool 
- ✓ Continue used the New MCP server splunk_run_query tool 

3. From this chat, you think that there are a couple of things that will really help you and your CFO in the future
 - a. Can you show me the Splunk searches used to generate the results in this chat?
 - b. Can you write a 500 word report based on these findings for my CFO?

Could you write a 500 word report based on these findings for my CFO

Let me create a comprehensive CFO report based on the findings:

Perfect! I've created a comprehensive 500-word CFO report based on all the findings. The report includes:

Report Contents:

1. **Executive Summary** - High-level overview of total spend and provider distribution
2. **Financial Overview** - Clear breakdown of the \$116.1M across three providers
3. **Account-Level Concentration Risk** - Highlights the GCP VM project concentration issue (\$32.6M)
4. **Service-Level Cost Drivers** - Shows that compute (44%), networking (26%), and databases (18%) drive the costs
5. **Key Findings & Recommendations** - Three critical findings about service concentration, volatility, and provider strategy

4. Finally, there are some stretch goals that might really impress your CFO:
 - a. How do you think you could automate some of this reporting?

- b. What about visualization of results?**
 - c. Exporting data to other tools would be helpful as well. How might you approach this?**
 - d. Were there any suggestions that your MCP client made as you were working through that you hadn't thought about?**
-